NSRIT

Semester End Examination, October / November, 2025

Degree	B. Tech. (U. G.)	**Program**	CSD & CSM			**Academic Year**	2025 – 2026
Course Code	23CS601	**Test Duration**	3 Hrs.	Max. Marks	70	**Semester**	V
Course	**COMPUTER NETWORKS**						

Part A (Short Answer Questions 5 x 2 = 10 Marks)

No.	Questions (1 through 5)	Learning Outcome (s)	DoK
1 a)	Describe unguided transmission media with two examples.	23CS601.1	L1
b)	Differentiate between Go-Back-N and Selective Repeat.	23CS601.2	L2
c)	What is the purpose of tunneling in networks	23CS601.3	L2
d)	What is congestion? What are the factors that cause congestion?	23CS601.4	L1
e)	Expand MIME and SMTP.	23CS601.5	L2

Part B (Long Answer Questions 5 x 12 = 60 Marks)

No.	Questions (6 through 15)	Marks	Learning Outcome (s)	DoK
2 (a)	Explain how does the OSI layer actually work in transmitting of data from one system to another system?	12M	23CS601.1	L2
	OR			
2 b) i	Explain any three network topologies used in the campus networking.	6M	23CS601.1	L2
ii	Compare Guided media and Un Guided media	6M	23CS601.1	L2
3 a	Explain about Cyclic Redundancy Check with an example?	12M	23CS601.2	L2
	OR			
3 b i	Explain about Sliding window protocol	6M	23CS601.2	L2
3 b ii	Explain token bus and token ring	6M	23CS601.2	L2
4 a i	Describe the hierarchical routing algorithm and illustrate it with an example.	6M	23CS601.3	L2
4 a ii	Differentiate between closed loop and open loop congestion control mechanisms.	6M	23CS601.3	L2
	OR			
4 b i	Illustrate the IPv4 header format and explain the significance of all the fields.	6M	23CS601.3	L2
4 b ii	What are the key differences between IPv4 and IPv6? Discuss their advantages and limitations	6M	23CS601.3	L2
5 a	Demonstrate how connection management is done in TCP	12M	23CS601.4	L2
	OR			
5 b I	Explain the three-way handshake mechanism used in TCP connection establishment.	6M	23CS601.4	L2
5 b ii	Explain Remote Procedure Call	6M	23CS601.4	L2
6 a i	Explain the resource records of DNS with its attributes	6M	23CS601.5	L2
6 a ii	Explain SMTP.	6M	23CS601.5	L2
	OR			
6 b	Explain Email Architecture	12M	23CS601.5	L2